

# Under five minutes, **ten slides.**

What to say on every slide, **29 September.**

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## Run of show · 29 Sept

Under five minutes, ten slides: 460 spoken words and 15 clicks, 3:38 at 150 words a minute. Detail that is not on a slide is in the notes here. The spine: **self, us, now** (02, 03), then **why, how, what** (04, 05, 06), then **who else, where it goes, the ask** (07–10). 07 and 08 are the fastest stretch: one beat each.

| SLIDE                            | CLOCK     | SEC | CLICKS | BEAT        | ONE LINE TO LAND                                    |
|----------------------------------|-----------|-----|--------|-------------|-----------------------------------------------------|
| 01 Control is an illusion        | 0:00–0:11 | 12  | 1      | hook        | Confidence without control.                         |
| 02 Founder                       | 0:11–0:39 | 28  | 2      | self        | This is the thing I kept needing and nobody sold.   |
| 03 Do my cells look normal?      | 0:39–1:02 | 22  | 1      | us, now     | Cells are the input nobody certifies.               |
| 04 One lab, or six?              | 1:02–1:34 | 33  | 3      | why<br>now  | The model isn't the hard part.<br>The reference is. |
| 05 One frame for every lab       | 1:34–2:10 | 36  | 4      | how         | One frame, and every lab that joins tightens it.    |
| 06 The product                   | 2:10–2:33 | 23  | 1      | what        | Send your controls. See where you stand.            |
| 07 Identity, yes. Behaviour, no. | 2:33–2:43 | 10  | 1      | who<br>else | The other half of the certificate.                  |

| SLIDE                               | CLOCK     | SEC | CLICKS | BEAT        | ONE LINE TO LAND                          |
|-------------------------------------|-----------|-----|--------|-------------|-------------------------------------------|
| 08 Three buyers, one certificate    | 2:43–2:58 | 15  | 1      | who buys    | Contributors use it free.                 |
| 09 The ugliest dataset in biology   | 2:58–3:13 | 15  | 1      | the build   | Name the failure, not just flag it.       |
| 10 Every plate, placed in the field | 3:13–3:33 | 20  | 0      | vision, ask | Have images of U2OS cells?<br>Talk to me. |

Captions stay hidden while you present (Tweaks › captions); the shared link and PDF show them. If you're late, skip the fourth view of 05 and the first sentence of 10.

## 01 Control is an illusion (0:00–0:11)

*On screen while you're introduced. Two lines drop from the top like dive lines: green is the field's datum, pink is yours, and the gap between them is the product. Let the dial drift.*

**0:00** Hi, I'm Kat. Every result in biology is a difference from a control, and nobody checks the control. So I'm building confidence without it. *Click.*

## 02 Founder (0:11–0:39)

*Self. Let the logos and the pink line under your name do the credential work.*

**0:11** Five labs, and the same question at every bench: do my cells look normal? Nobody could answer it. *Click.*

**0:20** At Axiom we onboarded hepatocytes through a good CRO. Their checks passed; ours passed. The first sign anything had moved came from outside: the Broad couldn't align our profiles to their reference.

**0:33** This is the thing I kept needing and nobody sold. *Click.*

**Founder detail, for the judge question (about 30 seconds).** Three methods, three papers, one thread. At AbbVie I built a neuronal tau aggregation assay for drug discovery [3]: the readout was what the cells did, not a label on them. At Axiom I built Cell Painting in primary human hepatocytes for cytotoxicity and mode of action [1], ran assay development for two of the largest Cell Painting datasets in the world, and invented two bioactivity detection methods that see a small molecule's effect on cells at about ten times lower concentration than LDH, ATP or cell count. With the Carpenter lab we showed that cell count alone predicts small-molecule bioactivity benchmarks [2]: even the simplest phenotype carries the signal. The thread is that bioactivity is a measurement of behaviour, and behaviour is the half of QC nobody measures. Every lab already images its control wells; Datum Line turns that image into the certificate, against the field.

**Precision to hold.** The question predates plates: at MGH it was whether a *Xenopus* oocyte, a single frog egg cell, looked normal before a recording. Say only what your author contribution statements support: "built" on [1] and [3], "we showed" on [2]. The two bioactivity methods and the ten-fold sensitivity are from your resume ([macrofluidic.me/resume](http://macrofluidic.me/resume)); if asked, name the comparison assays and the concentration range. The 2019 GABA<sub>A</sub> paper [5] is the MGH anesthesia lab, Harvard Medical School, 2017 to 2018. The industry claim is the deck's own: STR and vendor certificates cover identity, bioactivity covers behaviour, and behaviour is what a control well measures in any lab that images one.

**Precision to hold.** "Published with" lists coauthor institutions, never partners. The Axiom story has no villain: the CRO passed every check it had, Axiom passed every check it had, and the failure was that no check for this existed. Name no vendor and no person. The uneven distribution was noticed; its severity and its effect on data quality were unknown, so never present it as the cause, and never say "spun down unevenly". If asked what it was: "We still don't know for certain. The uneven spreading is one hypothesis; whole-well images show it more easily than our fields, and a field reference is what would tell you whether it matters." The sequence is: local checks passed, uneven spreading noticed but unquantified, the Broad flagged that reference

profiles would not align, the cause is still open. Say what happened, not how it was fixed. Confirm with Axiom that you can tell it; if not, say "at one lab".

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### 03 Do my cells look normal? (0:39–1:02)

*Us, then now. Point at the pink slice, then the six search results, then the two edges of card 3: run it, or hold.*

**0:39** It's not just me. About twenty-eight billion dollars a year of US preclinical research can't be reproduced, and a third of that is inputs.

**0:49** Today you search, you ask around, then you guess. I've seen one lab run too fast and one hold too long. Cells are the input nobody certifies. *Click.*

**Precision to hold.** \$28B is a 2015 US estimate; the 36% covers all reagents and reference materials, not only cells. The search card shows six real U2OS images from six papers, 2009 to 2023, via IDR and JUMP; sources are in the README. Card 3 names neither lab and gives no exact figure; keep it that way on stage. Growth chart is the metaphor if you need one: nobody plots their cells against the population.

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### 04 One lab, or six? (1:02–1:34, three views)

*Why now. The first view is a game: let the room look for four seconds.*

**1:02** Six untreated wells. Same line, same protocol, no drug. One lab, or six? *Click.*

**1:09** Six. Nobody here can tell. A classifier can: 99.9 percent across eleven labs, against nine by chance, on plates it never saw. *Click.*

**1:20** Why now: this data is public and free, and models are a commodity. The model isn't the hard part. The reference is. *Click.*

**Precision to hold.** Eleven labs, 93,228 untreated wells, 1,872 plates, held out by plate; 78 wells wrong. Holding out whole batches gives 99.6%; labs on one

microscope model, 99.99%. The signature is in every feature family, image-quality features included, so it reflects the whole imaging pipeline as well as the cells. Say "the lab", not "the cells". Cell Painting fluorescence; brightfield is the bet.

**For ML people.** The line on the second view: normalising each plate to its own controls takes lab identity from 99.9 to 24 percent. Right for discovery, wrong for QC (A3).

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## **05 One frame for every lab (1:34–2:10, four views)**

*How. Dark slide: slow down. Four views, one sentence each.*

**1:34** Eleven labs, same cells, same protocol: eleven different normals. That's what the classifier saw. *Click.*

**1:42** Put them in one frame, and the reference tightens with every lab that joins. *Click.*

**1:49** In that frame, answers move closer. The same drugs, measured in every lab, agree better: in ten of eleven labs, better than their own controls can do. *Click.*

**2:02** And where you sit predicts what you measure. So the map is worth having. *Click.*

**Figure detail.** First view: a PCA of 15,352 untreated wells for the eye; distances are measured in all features. The number is the mean distance between two labs' centres, 6.2 within-plate spreads. Second view: two references, each built from the same number of labs that share none; with one lab each they sit 6.2 apart, with five each 2.8. The dashed line is one over root  $n$ . A lab's own baseline settles sooner: 0.89 with one plate, 0.39 with eight, then flat. Third view: each batch's untreated wells are moved onto the field's mean and covariance, built without that lab, and the same move is applied to the treated wells. Agreement is the cosine between a batch's effect of 7 positive controls and the mean effect in the other labs: 0.75 as measured, 0.78 in the field's frame, higher in 121 of 125 batches. Normalised to the batch's own controls, the usual practice, 0.76; the frame is above that in 107 of 125 batches.

Fourth view: one point per batch; rho is Spearman after removing each lab's mean, minus 0.71 over 118 batches; shuffled, minus 0.03; without lab 2, minus 0.68; drift and effect from different wells, minus 0.72.

**Precision to hold.** The gain from the field's frame is consistent and modest: 0.75 as measured, 0.76 on own controls, 0.78 in the frame. It is between labs. A lab's agreement with its own first batch does not change (0.88, 0.88), and effect sizes vary more from batch to batch (0.06 becomes 0.12). Say "closer", not "the same". The margin over the wrong drug rises in 125 of 125 batches, so the frame is not making every drug look alike (A14). The ringed points on the fourth view are one lab's autumn batches (A13); the crosses are batches whose positive controls show no effect (A8). If you are late, skip the fourth view.

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## **06 Send your controls. See where you stand (2:10–2:33)**

*What. One sentence, then the map, then the certificate, then the row of ticks.*

**2:10** The product: send the control images you already take. They stay yours.

**2:15** Back comes your place in the field, a certificate you can show, and your data in the field's frame. Every batch you send makes the reference denser, for you and for everyone. It's a growth chart your cells come back to. *Click.*

**Precision to hold.** This is a replay on public data, not a customer. The same lab's later batches sit outside its baseline from 30 August, and the certificate says so and says where to look: ER and AGP stain intensity (A11, A12, A13). Across the field 6 of 129 batches are outside. Bring that up if asked whether it ever says no. Brightfield is still the bet (A10).

**Figure detail.** One real lab on one real day: lab 2, 16 August, 25 plates, chosen as its in-distribution batch with the most plates. 2.8 spreads from its own baseline, 4.4 to lab 6, 188 cells per well. The reference is 1,652 plates from the other 10 labs. In the field's frame this batch's agreement with the other labs goes from 0.70 to 0.77. The ticks are the batches sent so far; the open one is the next.

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## 07 Identity, yes. Behaviour, no. (2:33–2:43)

*Point at the open tick in the Datum Line column: identity is done with a partner, STR. One sentence.*

**2:33** STR proves which line you have. Nothing proves it behaves like everyone else's. STR is a collaborator, not a rival. *Click.*

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## 08 Three buyers, one certificate (2:43–2:58)

*Who buys. Walk the three columns; the pink row is what they pay.*

**2:43** Three buyers. Cores contribute controls and use it free: that builds the reference. Pharma and CROs pay per line, per quarter: a fraction of one unchecked screen. Cell banks pay per certified lot. *Click.*

**Precision to hold.** The prices are a proposal, not a validated model: free for contributors, about 5 to 15 thousand dollars per line per quarter for screening groups, a fee per certified lot. One unchecked screen is about 2.6 million, so the subscription is about 170 times less. Bad-lot alerts come after the certificate.

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## 09 The ugliest dataset in biology (2:58–3:13)

*The build. This is the blank-cheque answer, said before anyone asks. Point at the six real wells, then the four corrupted tiles.*

**2:58** With a blank cheque I'd build the ugliest dataset in biology, on purpose. Six labs already look this different. Then plant the failures, and make every method name which one went wrong. *Click.*

**Figure detail.** Six real untreated wells, one per lab, with each lab's instrument; across the 13 JUMP labs there are five microscope models, objective NA from 0.4 to 1.0, and eight plate types. The four red-team tiles are one real well corrupted in

software and are labelled simulated. The failures to plant: bubble, focus, exposure, clipped well, bad lot, wrong label.

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## **10 Every plate, placed in the field (3:13–3:33)**

*Vision, then the ask. Point at the open row, then the right card. Stop on the question.*

**3:13** The vision: every plate you image, placed in the field. Already in the frame, from public data: 129 batches, eleven labs. The next row is yours.

**3:24** What I need: images, U2OS first; ML people better than me; bench time; and a pre-seed.

**3:30** Have images of U2OS cells? Talk to me.

**Precision to hold.** Price and test-set budget are not set: say "per line, per quarter" and "the proof of concept sets the size". Don't name a partner lab until one has agreed. Name no JUMP lab beyond its number. The grid is 129 batches and 1,871 plates; each square is a batch. Now is U2OS in fluorescence; more lines and brightfield are next and are not yet shown.